

Yuvraj Kashyap

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Full-Stack Software Engineer

EDUCATION

University of Texas at Dallas

Expected May 2027

B.S. in Computer Science, Minor in Finance, Certificate in Entrepreneurship & Innovation

GPA: 3.77/4.00 | Dean's List | NCAA Division II Tennis

Relevant Coursework: Advanced Algorithms, Data Structures & Algorithms, Database Systems, Systems Programming, Software Engineering, Programming Languages, Machine Learning, Artificial Intelligence, Computer Vision, Linear Algebra, Probability & Statistics

EXPERIENCE

Software Engineering Intern

Incoming

Base Power

Austin, TX

Software Engineering Intern

Apr 2026 – Aug 2026

Medceptor – HealthTech startup

- Enabled expansion into two nursing markets (CNA and LPN) by shipping 14+ PRs on a five-person engineering team for core question-bank and exam workflows used by paying, recurring users.
- Enabled support for 100+ generated questions and automated import validation through Next.js, TypeScript, Django, and Supabase/Postgres workflows for track/level filtering, scoring, and review; carried features through testing and release QA.

Undergraduate Researcher, UAV / Smart City Systems

Oct 2025 – Present

University of Texas at Dallas

- Built a reproducible Python urban-geometry and UAV simulation framework from 1,553 OpenStreetMap building footprints across a 4 km × 4 km Dallas environment for path planning, visibility, coverage, and obstacle avoidance.
- Implemented 2.5D line-of-sight analysis, used greedy set cover to cover 224 of 240 sampled targets with six observers, and computed a 4.99 km eight-neighbor A* route around altitude-dependent obstacles.

Software Research Assistant, NOVA Autonomous Driving

Sep 2023 – Aug 2025

University of Texas at Dallas

- Reduced 3D sensor-data processing time with Python voxel-grid downsampling, cutting scan size by 94.5% (370,277 to 20,528 points) while preserving geometric structure.
- Debugged power, sensor, and I/O reliability issues across autonomous vehicle subsystems, translating hardware constraints into software and systems-level fixes for perception reliability.

Engineering Lead & Senior Consultant, Consult Your Community

Jan 2026 – Present

University of Texas at Dallas

- Delivered Glydr's web-based gaming hub prototype, adopted by the client and reaching 1,500+ users; improved latency and owned technical delivery from requirements and scoping through final handoff.

PROJECTS

Sticky – Full-Stack Task & Reminder Platform

TypeScript, Next.js, React, Hono, Supabase, PostgreSQL, TanStack Query, Zod, Vercel

- Deployed a private task-management PWA with recurring tasks, durable reminders, calendar planning, and realtime sync.
- Kept task writes and reminder events atomic with a transactional outbox; enforced scoped, idempotent commands through a versioned API and owner-level database access policies.

Atlas – Fault-Tolerant Distributed Crawl, Index & Search Platform

Python, FastAPI, PostgreSQL, Redis/RQ, OpenSearch, Docker, AWS, Terraform, React, Vite

- Recovered crawl and indexing jobs after worker failures using persistent scheduling, leases, retries, and outbox delivery.
- Passed a deterministic 10,000-page fault benchmark covering worker loss, complete Redis data loss, retries, duplicate delivery, redirects, malformed content, and transient failures.

JARVIS – Local AI Agent Runtime

Python, React, TypeScript, Rust, Tauri, SQLite, Pydantic, WebSockets, Playwright, Ollama

- Built a local agent runtime connecting durable memory, typed tools, and a desktop interface; enforced action permissions outside the model with explicit approvals, cancellation, and audit records.
- Validated tool execution, undo, replay denial, and backup recovery; passed 396 Python, TypeScript, and Rust tests. Daily-use qualification remains gated on local-model hardware capacity.

WorldState – GPU-Backed Real-Time Vision Runtime

Python, FastAPI, PyTorch, CUDA, OpenCV, WebSockets, PostgreSQL, React, Docker

- Decoupled live vision updates from slow model and storage work using asynchronous inference and shared-memory workers.
- Verified a 30-minute learned-model soak at 30.93 useful updates/s and 33.69 ms p95 with 2,588 MiB flat reserved VRAM, isolating storage latency from the frame-acknowledgement path.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Backend & APIs: FastAPI, Django, Hono, Node.js, REST APIs, Server Actions, SQLAlchemy, Alembic, Pydantic, Zod

Data, Search & Infrastructure: PostgreSQL, Redis/RQ, OpenSearch, Qdrant, Supabase Auth/RLS, Docker, Docker Compose, Git, GitHub Actions, Vercel, Doppler, Linux/Unix, CI/CD

AI/ML Systems: BM25, vector search, embeddings, cross-encoder reranking, retrieval evaluation, OpenRouter, Exa, LangChain, PyTorch, NumPy, Pandas, OpenCV

Frontend & Integrations: React, Next.js, TanStack Query, Tailwind CSS, Vite, Framer Motion, Recharts, Stripe, Resend, QStash